

```
#include <stdio.h>

#include <string.h>


#define MAX 100


// Stacks for BACK and FORWARD history
char backStack[MAX][50];
char forwardStack[MAX][50];


int backTop = -1, forwardTop = -1;


// Push page onto a stack
void push(char stack[][50], int *top, char page[]) {
    if(*top < MAX - 1) {
        (*top)++;
        strcpy(stack[*top], page);
    }
}


// Pop page from stack
void pop(char stack[][50], int *top, char page[]) {
    if(*top >= 0) {
        strcpy(page, stack[*top]);
        (*top)--;
    }
}
```

```
// MAIN PROGRAM
```

```
int main(){
```

```
    char currentPage[50] = "Home";
```

```
    int choice;
```

```
    char newPage[50], tempPage[50];
```

```
    while(1) {
```

```
        printf("\n--- Browser Navigation (STACK Simulation) ---\n");
```

```
        printf("Current Page: %s\n", currentPage);
```

```
        printf("1. Visit New Page\n");
```

```
        printf("2. Back\n");
```

```
        printf("3. Forward\n");
```

```
        printf("4. Exit\n");
```

```
        printf("Enter choice: ");
```

```
        scanf("%d", &choice);
```

```
        if(choice == 1) {
```

```
            printf("Enter new page: ");
```

```
            scanf("%s", newPage);
```

```
            push(backStack, &backTop, currentPage); // save current in BACK stack
```

```
            strcpy(currentPage, newPage);
```

```
            forwardTop = -1; // Clear forward history when new page visited
```

```
        } else if(choice == 2) {
```

```
            if(backTop == -1) {
```

```

        printf("No BACK history!\n");
    } else {
        push(forwardStack, &forwardTop, currentPage); // save current in FORWARD
        pop(backStack, &backTop, currentPage);    // move back
    }

} else if(choice == 3) {
    if(forwardTop == -1) {
        printf("No FORWARD history!\n");
    } else {
        push(backStack, &backTop, currentPage);    // save current in BACK
        pop(forwardStack, &forwardTop, currentPage); // move forward
    }

} else if(choice == 4) {
    printf("Exiting...\n");
    break;
}
else {
    printf("Invalid choice.\n");
}
}

return 0;
}

```

OUTPUT

--- Browser Navigation (STACK Simulation) --- Current Page: Home

1. Visit New Page
2. Back
3. Forward
4. Exit Enter choice: 1 Enter new page: Youtube

--- Browser Navigation (STACK Simulation) --- Current Page: Youtube

1. Visit New Page
2. Back
3. Forward
4. Exit Enter choice: 2

--- Browser Navigation (STACK Simulation) --- Current Page: Home

1. Visit New Page
2. Back
3. Forward
4. Exit Enter choice: 3

--- Browser Navigation (STACK Simulation) --- Current Page: Youtube

1. Visit New Page
2. Back
3. Forward
4. Exit Enter choice: 4 Exiting...

=== Code Execution Successful ===